Maloney's Geomethry

An Axiomatic Formalization of Synthetic Geometry

$$\mathfrak{M} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong, \mathcal{L}, \mathcal{A} \rangle$$

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Guide to Notation

Objects and Sets

A, B, C, $\ldots$, X, Y, Z
Capital italic: **points**

$\mathbf{P}$ The set of all points
$\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\ldots \mathfrak{X}$, $\mathfrak{Y}$, $\mathfrak{Z}$ Capital x: geometric figures
$\overline{AB}$ **Line Segment**
$\overleftrightarrow{AB}$ **Line**
$\overrightarrow{AB}$ **Ray**
$\triangle ABC$ **Triangle**
$\angle ABC$ **Angle**
$\overline{AB} \parallel \overline{CD}$ **Parallel**
$\overline{AB} \perp \overline{CD}$ **Perpendicular**

0.1 Primative Relations

$\mathcal{B}(A, B, C)$ Betweeness
$A = B$ Equivalence
$\overline{AB}$, $\overline{CD}$ Congruence

0.2 idk

$\mathcal{L}$ The **Language**
$\mathcal{A}$ The **Axiom Set**

Part I

Philosophical Foundations

Chapter 1

The Ontology of Geometry

1.1 Existence: The Reality of Mathematical Objects

Is math real? Discussing Platonism vs. Nominalism.

1.2 Origin: Discovery and Invention

1.3 Modality: Necessity, Possibility, and Contingency

1.4 Identity: The Nature of Objects and Relations

Chapter 2

Epistemology of Mathematics

2.1 The Nature of Truth

2.2 Axioms, Theorems, Structure

2.3 Definitions

2.4 Proof, Justification, and Rigor

Part II

Logical Foundations

Chapter 3

First-Order Logic

3.1 Logical Syntax

3.2 Logical Connectives

3.3 Variables and Quantifiers

3.4 Rules of Inference and Deduction

Chapter 4

The Signature of Geometry

4.1 Formal Signatures

Specification of primitive symbols and arities defining a formal system.

Definition 1 (Signature of Geometry). *A signature $\mathcal{L}_{geo}$ consists of:*

- *a ternary relation symbol $\mathcal{B}$ (betweenness),*
- *a quaternary relation symbol $\cong$ (congruence).*

4.2 Geometric Language

Formal language generated from geometric primitives and relation symbols.

4.3 Typed Signature Extension

Signatures enriched with dependent types to encode geometric constraints directly in syntax.

4.4 The Geometric Language and Well-Formed Formulas

Chapter 5

The Axiomatic System

5.1 Primative Objects

5.2 Primative Relations

5.3 Axioms

5.4 Derived Notions

5.5 Derived Proofs

5.6 Models and Structures

Part III

Set-Theoretic Geometry

Chapter 6

Primative Notions

For our axiomatic system, we define three primative notions comprising of one object and two relations that together form a structure. The first of these are points, which are designnated using upper case latin letters, $A, B, C, \ldots$. Each point is a member of the set **PP**, which is defined as $\mathbf{P} \coloneqq \{P \mid Pisapoint\}$. This set is equpped with a ternary relation $\mathcal{B}$ (betwenessness) and an equivalence relation $\cong$ (congruency). Together, these form the plane $\mathcal{P}$.

$$\mathcal{P} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong \rangle.$$

Let the set $\mathbf{P}$ denote the set of all points such that

$$P \in \mathbf{P}.$$

6.1 Plane

Let the plane there be a structure $\mathcal{P}$ equipped with a domain $\mathbf{P}$ and relations $\mathcal{B}$ and $\cong$.

Definition 2 (Plane)**.** *A plane is a structure*

$$\mathcal{P} = (P, \mathcal{B}, \cong)$$

where P is a non-empty domain.

6.2 Points

The domain of the plane, denoted $\mathbf{P}$, is referred to as the set of all points. In this foundational layer, a "point" is defined not through simplier parts, but rather solely by its membership in this set. That is, an object P is a point if $P \in \mathbf{P}$. Points are denoted using capital italics such as A, B, C, $\ldots$.

Intuitvely, a point can be thought of as a unique position in space/on the plane, or in the Euclidean tradition a point is "that which has no part". A point has no magnitude, length, width, or depth, and thus can be considered (informally) as 0-dimensional. However, in our formal system we move away from physical descriptions and treat points simply as a primative object that cannot be defined further. Formally, we define a point as follows:

$$\forall P (P \text{ is a point} \iff P \in \mathbf{P}).$$

6.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. We write $\mathcal{B}(A, B, C)$ to assert that point B lies *between* points A and C.

Unlike a set $\{A, B, C\}$, where the order of elements is irrelevant, the relation $\mathcal{B}$ is defined as the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus,

$$\mathcal{B} \subseteq \mathbf{P}^3 := \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

Saying that the betweness relation holds for A, B and C is equivlalent to saying that their triple is an element of that subset:

$$\mathcal{B}(A, B, C) \iff (A, B, C) \in \mathcal{B}.$$

6.4 Congruence

Congruence is a quaternary relation on $\mathbf{P}$ used to denote that two pairs of points are equidistant. While typically visualized as a binary relation between segments, in our point-primitive system, it is formally an equivalence relation on the set of ordered pairs $\mathbf{P} \times \mathbf{P}$. We denote this relation using the symbol $\cong$.

If the relation holds for the pairs (A, B) and (C, D), we write $(A, B) \cong (C, D)$, which is interpreted as the distance from A to B being equal to the distance from C to D.

Intuitively, congruence represents the "rigidity" of the plane. It allows us to move geometric configurations from one location to another while preserving their size and shape. In the Euclidean tradition, this corresponds to the ability to place one segment onto another such that they coincide perfectly. Formally, this relation satisfies the requirements of an equivalence relation—reflexivity, symmetry, and transitivity—on the domain of point-pairs.

Chapter 7

Axioms

7.1 Axioms of Equality

Axiom 1 (Idenity/Reflexivity of Equality)**.**

$$\forall A \in \mathbf{P}, A = A.$$

Axiom 2 (Symmetry of Equality)**.**

$$\forall A, B \in \mathbf{P}, A = B \iff B = A.$$

Axiom 3 (Transitivity of Equality)**.**

$$\forall A, B, C \in \mathbf{P}, A = B \land B = C \implies A = C.$$

7.2 Axioms of Betwneenness

The betweeness relation is governed by the axioms of betweeness.

Axiom 4 (Idenity/Reflexivity of Betweenness)**.**

$$\forall A, B \in \mathbf{P}, \mathcal{B}(A, B, A) \iff A = B.$$

Axiom 5 (Symmetry of Betweenness)**.**

$$\forall A, B, C \in \mathbf{P}, \mathcal{B}(A, B, C) \iff \mathcal{B}(C, B, A).$$

Axiom 6 (Order of Betweeness)**.**

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (\neg\mathcal{B}(B, A, X) \land \neg\mathcal{B}(A, X, B)).$$

Axiom 7 (Inner Point/Density of Betweeness)**.**

$$\forall A, C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

Axiom 8 (Outer Point/Extensibility of Betweenness)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \exists C \in \mathbf{P} \colon \mathcal{B}(A, B, C).$$

7.3 Axioms of Congruence

$$\mathcal{S} \coloneqq \{S \subseteq \mathbf{P} \mid 1 \leq |S| \leq 2\}.$$
$$\cong \subseteq \mathcal{S} \times \mathcal{S}.$$

Axiom 9 (Reflexivity of Congruence)**.**

$$\forall S \in \mathcal{S}, S \cong S.$$

Axiom 10 (Symmetry of Congruence)**.**

$$\forall S_1, S_2 \in \mathcal{S}, S_1 \cong S_2 \iff S_2 \cong S_1.$$

Axiom 11 (Transitivity of Congruence)**.**

$$\forall S_1, S_2, S_3 \in \mathcal{S}, (S_1 \cong S_2 \wedge S_2 \cong S_3) \implies S_1 \cong S_3.$$

Axiom 12 (Idenity of Indiscernibles)**.**

$$\forall S_1, S_2 \in \mathcal{S}, (|S_1| = 1 \wedge S_1 \cong S_2) \implies |S_2| = 1.$$

7.4 idk

The ruler axiom states that given a point-pair S and two discint points A and B, there exists one point X such that $\mathcal{B}(A, B, X)$ and $\{B, X\}$ is congruent to S.

Axiom 13 (Ruler Axiom)**.**

$$\forall S \in \mathcal{S}, \forall A, B \in \mathbf{P}, A \neq B \implies \exists! X \in \mathbf{P} \colon \mathcal{B}(A, B, X) \wedge \{B, X\} \cong S.$$

Chapter 8

Partition of the Plane

8.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a parition splitting the plane into two discint, non-overlapping regions, which is then used to describe a nototion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 14 (Plane Separation)**.** *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \wedge H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l\colon\ (\exists X \in l, \mathcal{B}(A, X, B) \iff \{A, B\} \not\subset H_{l1} \wedge \{A, B\} \not\subset H_{l2})$.

explain the meaning-of-axioms-here

8.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a seperation/parition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l. This definition ensures the relation is based upon the primative notion of betweeness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 3 (Same-Side Relation)**.** *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l\colon \mathcal{B}(A, X, B).$$

8.3 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane ($H_{l,A}$). This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 4 (Half-Plane)**.** *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane detirmined by l and A, denoted $H_{l,A}$, is the subset of the point-set consistening of all points on the same side of l as A. That is,*

$$H_{l,A} \coloneqq \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same same as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibily useful in defining derived geometric figures.

8.4 Theorems

Theorem 1 (Same-Side Reflexivity)**.** *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P} \colon \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l \colon \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfing these conditions. As a consequence, by the Idenity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. □

Theorem 2 (Same-Side Symmetry)**.**

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l \colon \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweeness, the right-hand side of this is equivalent to $\nexists X \in l \colon \mathcal{B}(B, X, A)$, thus showing

$$X \in l \colon \neg\mathcal{B}(A, X, B) \iff X \in l \colon \neg\mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. □

Theorem 3 (Same-Side Transitivity)**.** *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Seperation Axiom, we know there must exist two distint sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B, and C are not on l, each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarily, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane seperation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. □

randomideas:

Chapter 9

Angles

As opposed to being a measure, we define an angle as angle as a geometric figure, ie: a set of points. Specifically, an angle is the union of two rays that share a common vertex.

Definition 5 (Angle)**.** *An* ***angle*** *is the union of two rays $\overrightarrow{AO}$ and $\overrightarrow{OB}$ that share a common vertex O, denoted as follows*

$$\angle AOB \coloneqq \overrightarrow{AO} \cup \overrightarrow{OB}.$$

Using the half-planes defined previously, we have the tools neccessary to describe the interior and exterior of an angle.

Definition 6 (Interior of an Angle)**.** *For any three non-collinear points $A, B, C \in \mathbf{P}$, the* ***interior*** *of $\angle ABC$ is the set:*

$$Int(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$Int(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Definition 7 (Interior of an Angle)**.** *For any three non-collinear points $A, B, C \in \mathbf{P}$, the* ***interior*** *of $\angle ABC$ is the set:*

$$Int(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$Int(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Chapter 10

Linear Notions

10.1 Defintion of Linear Notions

A segment is a subset of $\mathbf{P}$ defined as the union of two discint points and all points between them. Symbolically, that is

Definition 8 (Segment)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

A ray pointing from point A to a discint point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A, B, all points between A and B, and all points such that B is between A and that point, that is

Definition 9 (Ray)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B, all points between A and B, all points where A is between it and B, and all points where B is between it and A. That is,

Definition 10 (Line)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

10.2 Properties of Linear Notions

Theorem 4 (Distinctness of Betweenness)**.** *If $\mathcal{B}(A, B, X)$ holds, then A, B, and X are three distinct points:*

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (A \neq B \wedge B \neq X \wedge A \neq X).$$

Proof. We proceed by contradiction. Suppose $\mathcal{B}(A, B, X)$ holds.

First, assume $A = X$. Substituting A for X in our hypothesis, we obtain $\mathcal{B}(A, B, A)$, which implies $A = B$. If $A = B$, then the relation $\mathcal{B}(A, B, X)$ is equivalent to $\mathcal{B}(B, A, X)$. However, by the Order of Betweeness, $\mathcal{B}(A, B, X) \implies \neg\mathcal{B}(B, A, X)$. Here, we reach a contradiction and conclude our hypothesis must be false and thus $A \neq X$.

Next, we test the assumption $A = B$. Under this assumption, we substite our hypothesis to obtain $\mathcal{B}(B, A, X)$. Similarly, the Order of Betweeness shows these cannot be simutanously true, and thus our assumption must be false, meaning $A \neq B$.

Finally, by the Transitivity of Equality, $B \neq X$. Thus proving the result. $\square$

Theorem 5 (Segment Symmetry)**.** *For any two points A, B, the segment AB is equal to the segment BA.*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. □

Theorem 6 (Line Symmetry)**.** *For any two distinct points A and B, the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweeness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the by the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$ □

Theorem 7 (Ray Asymmetry)**.** *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overleftarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the defintion of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweeness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweeness we have $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(A, Y, B)$, which by the Symmetry of Betweeness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. □

Theorem 8 (Union of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. □

Theorem 9 (Intersection of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$ and all points X betweem A and B, that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB, as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB. □

10.3 Parallelism

Two lines are parallel if they have no point in common. In our set-theoretic foundations, this is to mean that the intersection of the lines is the empty set. Because we define parallelism as a binary relation on lines, we first must have a set of all lines to work with. Thus, we let $\mathbf{P}$ be the domain called the set of all points and define $\mathbf{L}$ to be the collection of all subsets of $\mathbf{P}$ that consist of the line $\overleftarrow{AB}$ for a any given two points $A, B \in \mathbf{P}$. Formally, that is:

$$\mathbf{L} \coloneqq \{l \subseteq \mathbf{P} \mid \exists A, B \in \mathbf{P}(A \neq B \wedge l = \overleftrightarrow{AB})\}.$$

Definition 11 (Parallel)**.** *Given any lines $l, m \in \mathbf{L}$, we say l is parallel to m and write $l \parallel m$ if and only if they are the same same set of points or their set intersection is empty.*

$$\forall l, m \in \mathbf{L},\ l \parallel m \iff (l = m) \vee (l \cap m = \emptyset).$$

add more here

Chapter 11

Triangles

11.1 Defintion

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Chapter 12

Quadrilaterals

motivation

12.1 Definition

A **quadrilateral is defined as the union of four unique noncollinear segments as follows:**

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each line segment is called a side of the quadrilateral, and each point A, B, C, D **is a vertex**

Theorem 10. *idk*

i hate this project lmao ts straight retarded. idk what to do with it ever.

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